

**Psychologically Harmful Caregiving, Emotional Abuse and Emotional Neglect: Differential
Associations with Inhibitory Control in Indian Emerging Adults**

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Title of the Proposed Study: Psychologically Harmful Caregiving, Emotional Abuse and Emotional Neglect: Differential Associations with Inhibitory Control in Indian Emerging Adults

Abstract

Thoughtful parenting is crucial for a child's social, emotional, cognitive development and is shaped by its social and cultural context. However, globally, almost six in ten children experience physical and/or psychological violence from their caregivers. In India, emotional abuse affects almost half of all children. Despite being highly prevalent, psychological maltreatment is under-recognized. Emotionally abusive actions that include humiliation, constant criticism and conditional affection are often normalised in the Indian sociocultural context. Moreover, most empirical studies from this domain use samples from Western populations with poor generalizability.

A consistent finding in the existing literature is that childhood maltreatment is linked with poorer executive functioning. Nevertheless, some key limitations still exist. Firstly, many researchers consider childhood maltreatment as a one-dimensional construct and do not distinguish between different types of adversity despite the fact that contemporary developmental models imply that different experiences affect neurodevelopment via different pathways. In particular, emotional abuse and emotional neglect have not been studied separately. Secondly, prior studies mainly employ emotionally neutral cognitive tests, which may not capture the influence of affect on inhibitory control. Research into the impact of childhood maltreatment on inhibitory control has also yielded inconsistent findings, pointing towards the need to conduct more theoretically grounded experiments.

The proposed research is an attempt to fill those gaps through the exploration of inhibitory control as the cognitive mechanism behind the long-term consequences of psychologically harmful caregiving, particularly childhood emotional abuse and emotional neglect. Based on the ideas presented in the Dimensional Model of Adversity and Psychopathology, this research interprets emotional abuse as a threat-based experience and emotional neglect as a deprivation-based experience that have different influences on inhibitory

control. The study also integrates Indian philosophical concepts of Manas, Buddhi, and Samskara with contemporary theories of executive functioning to provide a culturally grounded framework for understanding cognitive development.

The study will use a cross-sectional, correlational design with about 150 Indian participants between 18 and 25 years of age. The CTQ-SF will be used to measure childhood trauma and the DASS-21 to assess current psychological distress. In this study, emotional abuse and emotional neglect will be conceptualized as continuous predictors as opposed to discrete group categories. Dependent behavioral measures will comprise commission errors, omission errors, reaction times, reaction time variability, and signal detection sensitivity (d'). Associations between childhood psychological maltreatment, inhibitory control, and co-occurring types of maltreatment and psychological distress will be assessed using linear mixed-effects models.

It is predicted that emotional neglect will correlate with global deficits in inhibitory control regardless of the emotional context whereas emotional abuse will affect inhibitory control only when emotionally salient distractors are present. This study will distinguish between emotional abuse and emotional neglect and assess their associations within a single experimental paradigm in an Indian emerging adult population. This study seeks to reconcile conflicting results in previous research and shed light on the cognitive processes affected by childhood psychological maltreatment. The results of the study are expected to inform culturally relevant theories of Indian parenting, childhood adversity, as well as education and mental health interventions for prevention and early intervention in India.

Significance of Proposed Research

A child's social, emotional and cognitive development is strongly affected by the kind of parenting practices and environments they experience. This is shaped by a unique cultural context in India, which includes strong collectivist values, family interdependence, respect for elders and high parental involvement. Indian parenting involves high levels of warmth but also high expectations and standards for discipline [1,2,3]. These conditions are often helpful in promoting adaptive and healthy development but we know little about how these parenting styles influence development of children's cognitive functions. However, most of our understanding in this area is based on Western frameworks which do not emphasize familial responsibility, academic achievement and social acceptance.

Six in ten children under five years of age experience physical or psychological violence from their caregivers (WHO). In India, the prevalence of all forms of child abuse is alarmingly high with 1 in 2 individuals reporting adverse childhood experiences [physical abuse (32%), and emotional abuse (40%)] [4]. These experiences increase the risk of depression, suicide, substance abuse, numerous physical health conditions, and other negative outcomes [5]. Despite this, only a few studies in India have employed robust methodology and standardized instruments to study the long-term effects of parenting and childhood maltreatment. Consequently, a lot of what we know is based on western findings and is not generalizable to Indian societies.

Moreover, due to stigma and other social influences, emotional abuse and childhood adversity overall is underreported [6]. Behaviors including public shaming, conditional affection, humiliation, etc are often seen as normal or tough love. This normalization helps hide both its prevalence and effects [7]. This makes it essential to consider how well-intended parenting practices may end up psychologically harmful. Rather than viewing parenting and psychological maltreatment as separate constructs, this study conceptualises psychologically harmful caregiving as occurring at the maladaptive end of a broader continuum of parenting behaviours. Furthermore, psychological maltreatment is often viewed as secondary to physical and sexual forms even though it has been associated with psychiatric and functional impairments that were comparable to and in some cases even greater than those associated with physical or sexual abuse [8].

We know that adverse caregiving experiences have lasting effects on cognition, including executive function, attention, academic performance, memory, and so on [9,10,11]. However, there is no consensus on the specific mechanisms and pathways that may lead to such deficits. Moreover, many studies report mixed findings on how and some even report contradictory effects, indicating the need for a more nuanced approach. Most studies view childhood maltreatment as a single construct even though we now know that different forms of harmful caregiving may lead to different effects [12].

One mechanism that may explain the effects of harmful caregiving is inhibitory control or the ability to suppress unwanted thoughts, emotions, and actions, enabling individuals to maintain top-down attentional control on relevant tasks and filter out distractions. Inhibitory control influences learning, academic performance, decision-making, emotion regulation and social adaptation. It is also important for moral behavior, rule following, planning, deliberation, etc [13]. Since inhibitory control affects multiple cognitive processes, it may help us explain many impacts of psychologically harmful parenting practices including those in working memory, academic performance and attention. Understanding more about such a crucial cognitive function may enable us to formulate better parenting strategies, early interventions, and trauma informed educational and mental health practices.

Theoretical Framework and Scope of the Proposed Study

Beyond its significance in contemporary psychology, these constructs and theories closely align with what Indian philosophical texts have told us for centuries. Indian philosophy has largely viewed human cognition as consisting of multiple faculties while remaining deeply interconnected and codependent, especially the Yoga and Vedanta schools. Manas is the part of our cognition that controls the processing of sensory information and how our attention is allocated to them. Whereas, Buddhi represents the higher order discrimination, self regulation and judgement functions, which can be linked to inhibitory control and more broadly a top-down control of attention [14]. Together, these aspects provide us a framework to study attentional processes. Indian philosophies also have a lot to say about experiences and their effects on our cognitive development. Experiences in early life leave lasting effects or samskaras on our cognition [15]. The Yoga Sutras expand on this understanding by viewing cognitive well-being as a state of mind absent of fluctuations [16]. We can connect this and analyse how these samskaras of early childhood may create lasting fluctuations, leading to a need for improvement in cognitive control. While these concepts are from philosophical schools, they are similar to what contemporary science tells us.

These philosophical concepts are also reflected in how parents raise children in India. Traditionally, Indian parenting gives importance to interdependence, respect for elders, emotional connectedness, and the development of self discipline and responsibility within the family. Instead of prioritising autonomy and independence, children are socialized using deep relationships, guided participation, and moral instruction. Although Indian parenting has evolved recently from the influences of urbanization and globalization, these collectivistic values continue to exist. This cultural context is an important lens to understand how early caregiving experiences may shape the development of attentional control and self-regulation, making parenting an important pathway in linking early experiences to executive function development.

Executive function frameworks conceptualize goal directed behavior to be controlled by interconnected higher order functions which include inhibitory control, working memory and cognitive flexibility. They come together to enable important capacities such as planning, reasoning, problem solving and more. In the proposed study, we focus on inhibitory control which is crucial to resist distraction, maintain attentional focus and regulate behavior to fulfill internal goals. Inhibitory control is viewed as the foundation on which many of these functions depend upon. The ability to inhibit irrelevant information and competing response precedes others as our mental and physical resources are limited. For example, one must inhibit distracting sensory information to effectively maintain task relevant details in working memory [17]. This way, all of the executive functions are interconnected and it makes inhibitory control an especially important mechanism to study.

Developmental theories view executive functioning as a system that is dynamic and as one that changes both due to biologically driven processes and environmental experiences. Parenting influences executive function through several mechanisms including emotional responsiveness, cognitive stimulation, scaffolding, autonomy support and opportunities for self-regulation. Inhibitory control, in specific, goes through significant change during childhood. Children become better at suppressing prepotent responses, distractions, regulating emotional responses and adapting behavior to different environments over time. As there is high neural plasticity during this developmental phase, the quality of early childhood experiences and caregiving may be crucial in the maturation of these systems and processes [17,18]. Subsequently, variations in parenting practices may lead to different developmental trajectories and lead to long-term effects.

Within this cultural context, however, certain parenting practices intended to promote discipline, family obligation, or emotional restraint may, when persistent or extreme, inadvertently manifest as psychologically harmful caregiving [19,3]. The psychological maltreatment framework allows us to gain a more nuanced understanding of caregiving in India since harsh criticism, emotional invalidation, excessive expectations, or parentification are sometimes normalized in families. This allows us to understand childhood adversity beyond sexual and physical forms of abuse. Persistent patterns of caregiving that makes a child feel unwanted, unloved, unsafe or unvalued are characterized under this framework. This includes emotional abuse (humiliation, rejection, intimidation, criticism) and emotional neglect (emotional unavailability, lack of responsiveness, failure to meet emotional needs). Though neglect and abuse often occur simultaneously, they are found to have varying effects on cognition. Abuse leads to a threatening environment and may lead to heightened vigilance towards emotionally salient information, especially on negative emotions. Neglect leads to deprivation of emotional and cognitive stimulation which may inhibit the maturation of these executive functions.

Moreover, issues like parentification that are commonly encountered in India are still understudied [20]. Looking at these forms of emotional neglect may lead to better insights specific to our cultural context. This framework also complements our focus on inhibitory control as there is strong evidence that neutral and emotionally valent distractors affect inhibitory control differently [21]. By studying this mechanism from the psychological maltreatment perspective, we may be better able to explain the mixed findings in current literature and pinpoint underlying mechanisms that affect attention in more detail.

Review of Literature

Threat vs Deprivation

Within the broader literature on caregiving and child development, one of the most important advances has been the recognition that adverse caregiving experiences cannot be understood as a single, homogeneous construct [22]. Earlier studies typically combined diverse experiences of adversity including physical, sexual, and psychological abuse and neglect into cumulative maltreatment scores. Although this approach established robust associations between adverse caregiving experiences and poorer cognitive outcomes, it could not explain why different studies reported varying cognitive profiles or why certain domains appeared to be disproportionately affected. In the past, most studies combined diverse experiences of adversity (physical, sexual, psychological, abuse, neglect, etc), into cumulative “maltreatment” scores. This approach has still shown us that there are strong links between adversity and poorer cognitive outcomes. However, this does not explain why the findings vary significantly among these studies or why some of these cognitive domains appear to be influenced more than the others.

To address these limitations, McLaughlin and colleagues have proposed the Dimensional Model of Adversity and Psychopathology. Rather than treating caregiving adversity as a single construct, the model differentiates between experiences of threat and deprivation. Threat is defined as experiences that involve actual or threatened harm and deprivation as the absence of expected cognitive, social and emotional inputs. They propose that threat primarily influences emotional learning and processing salient information while deprivation seems to disrupt executive functions by limiting the environmental stimulation that is necessary for the maturation of prefrontal systems [23]. In a subsequent study, 169 adolescents (age 13–17) completed measures of childhood adversity, including neglect, abuse and community violence, alongside the Behavior Rating Inventory of Executive Function . After statistically accounting for both adversity dimensions, deprivation predicted greater executive function difficulties, whereas threat showed no independent association. These findings provided the initial behavioural and neurobiological evidence supporting the prediction that deprivation has a stronger influence on executive control than threat [10].

Support for this distinction was substantially strengthened by the systematic review and meta-analysis conducted by Dylan Johnson and colleagues. They put together 91 studies representing 82 different cohorts and 31,188 participants. Here, they examined if executive functioning varied based on the threat-deprivation model. When childhood adversity was considered a unitary construct, they discovered it to be associated with poorer cognitive flexibility, inhibitory control and working memory. However, when they analysed threat and deprivation based experiences separately, they found that deprivation showed stronger associations with inhibitory control (Hedges' $g = -0.43$) and working memory (Hedges' $g = -0.54$). Threat based experiences only showed a weaker association (Hedges' $g = -0.27$ and -0.28 respectively). These findings provide large-scale quantitative support for the central prediction of the dimensional model. Although both forms of adversity impair executive functioning, deprivation appears to have a disproportionately greater impact on inhibitory control and working memory [24].

More recently, Marina Schäfer and colleagues (2023) extended this framework using a large Brazilian cohort, addressing concerns that previous evidence had largely been restricted to high-income Western countries. Their findings replicated the model's core predictions: deprivation was more strongly associated with executive function deficits, whereas threat showed stronger relationships with emotional processing and psychopathology. Importantly, the authors argued that examining adversity dimensionally provides greater explanatory power than cumulative risk models and may improve our understanding of developmental outcomes across diverse sociocultural settings [25]. These findings suggest that examining specific dimensions of adverse caregiving may be particularly valuable in culturally diverse settings, where parenting practices and child-rearing norms vary considerably.

Childhood Maltreatment and Affective Inhibitory Control

Although executive function deficits following adverse caregiving experiences are well established, considerably fewer studies have examined inhibitory control under emotionally salient conditions, often referred to as affective inhibitory control. Unlike conventional inhibitory control tasks that use emotionally neutral stimuli, affective paradigms require individuals to inhibit responses while ignoring emotionally salient but task-irrelevant information. Because emotional stimuli compete more strongly for attentional resources, these paradigms may better capture the cognitive consequences of childhood adversity, particularly experiences involving interpersonal threat.

One of the earliest studies to investigate this question was conducted by Cohen-Gilbert et al., who developed an Emotional Go/No-Go paradigm using emotionally valenced background images while participants responded to centrally presented letters. Studying adolescents and young adults, they demonstrated that negative emotional distractors selectively impaired

inhibitory control compared with neutral distractors, particularly during early adolescence, indicating that emotionally salient information places greater demands on executive control than neutral contexts. Their findings established the Emotional Go/No-Go paradigm as a sensitive measure of interactions between emotion and inhibitory control and provided the methodological foundation for later studies examining childhood adversity [21].

Building directly on this paradigm, Lauren A. Demers and colleagues examined 38 adults with prospectively documented childhood adversity histories and 34 matched comparison participants while they completed an Emotional Go/No-Go task during functional magnetic resonance imaging (fMRI). Participants responded to letters presented over neutral or negative International Affective Picture System (IAPS) images. Contrary to what they expected, the comparison group performed worse in the Go/No-Go task with a poorer No-Go accuracy under negative compared with neutral background. However, the maltreated group did not deviate significantly during negative emotional distraction [25]. These findings contrast with those reported by Gray et al. (2016), who found that adolescents with documented histories of childhood adversity showed greater attentional interference from threatening stimuli, particularly angry facial expressions, suggesting heightened difficulty disengaging attention from emotionally salient information [27].

A similar approach was taken by Sætren et al., who examined affective inhibitory control in a nationally representative sample of adolescents. They used an Emotional Go/No-Go task with angry and neutral facial distractors. Instead of focusing on behavioral group differences alone, they examined whether affective inhibitory control can explain the link between childhood maltreatment and internalizing symptoms [28]. The participants made more commission errors when encountered with angry faces than neutral faces. This confirms that threatening emotional stimuli interferes with inhibitory control more than neutral stimuli. More importantly, this poorer affective inhibitory control was found to moderate the association between psychological abuse and internalising symptoms among girls. This suggests that the difficulty in inhibiting responses to threat related stimuli may point to an important mechanism linking childhood abuse and emotional difficulties later in life.

Research Gaps

Though the current literature provides substantial evidence that psychologically harmful caregiving is linked with poorer executive functioning, there are still many conceptual and methodological gaps that remain. These limitations have made it difficult to pinpoint the specific developmental pathways that link harmful caregiving and its influences on cognition.

First, although parenting encompasses a broad range of caregiving experiences, most of the existing research still conceptualizes childhood maltreatment as a single construct. Many

studies combine physical, sexual, emotional, neglect and abuse scores into one overall maltreatment score. This approach assumes that qualitatively different experiences lead to very similar impacts on similar mechanisms. As discussed previously, contemporary models such as the Dimensional Model of Adversity and Psychopathology challenge the assumption by arguing that different experiences may influence neurodevelopment differently. Recent empirical evidence also seems to support this. However, only a few studies investigate emotional abuse and neglect independently while also controlling for other forms of maltreatment. Therefore, it is still unclear if previously reported executive function deficits indicate the general effects of adversity or the specific influences of particular forms of psychological maltreatment.

Second, research investigating inhibitory control following adverse caregiving has relied predominantly on emotionally neutral cognitive paradigms. Traditional Go/No-Go, Stroop, and Flanker tasks provide valuable measures of executive functioning but may underestimate the influence of emotionally relevant information on cognitive control. Evidence from affective inhibitory control paradigms demonstrates that emotional distractors compete more strongly for attentional resources than neutral stimuli. Moreover, threat-based experiences are theorized to alter emotional salience processing rather than executive functioning globally. Despite these theoretical developments, very few studies have manipulated emotional context while simultaneously distinguishing between emotional abuse and emotional neglect.

Third, existing findings regarding affective inhibitory control following childhood maltreatment remain inconsistent. Though some studies find threatening stimuli to interfere with inhibitory control in maltreated individuals, some other studies find them to be better at accounting for these distractors. These contradictory findings may be a result of genuine differences in developmental processes but they may also be because of the difference in operationalization of childhood adversity, task paradigms, or due to treating childhood maltreatment as a single construct. Very few studies have tried to compare competing theoretical explanations using the same experimental design. Therefore, the field is still lacking evidence that can determine whether the inconsistent findings are due to methodological differences or due to fundamentally different cognitive mechanisms associated with different forms of adversity.

Fourth, there is a notable lack of research examining these questions within Indian populations. The overwhelming majority of empirical work on childhood maltreatment and executive functioning has been conducted in North American and European contexts. However, emotional abuse and emotional neglect often occur within distinct sociocultural environments. In India, psychologically harmful parenting practices including humiliation, excessive criticism, conditional affection, emotional invalidation, and harsh disciplinary practices may be more commonly accepted. Such cultural norms may influence both the prevalence of psychological maltreatment and the ways in which individuals process and respond to emotionally salient information later in life. Despite these contextual factors, there is little experimental evidence

investigating the cognitive consequences of psychological maltreatment among Indians using standardised paradigms.

The proposed study addresses these gaps by examining emotional abuse and emotional neglect as distinct forms of psychologically harmful caregiving within the broader context of parenting. It employs an Emotional Go/No-Go paradigm across multiple emotional contexts, treats childhood experiences dimensionally rather than categorically, controls for co-occurring maltreatment and current psychological distress, and investigates these relationships in an Indian emerging adult sample. In doing so, the study aims to provide a more theoretically grounded and culturally contextualised understanding of how early caregiving experiences shape inhibitory control.

Objectives of the Proposed Study

Existing literature has shown us that psychologically harmful caregiving is associated with harmful psychological and developmental impacts. However, we know little about the specific mechanisms that connect these early childhood experiences with functioning in adult life, especially among non-clinical populations. The proposed research aims to take a novel and nuanced approach by considering specific forms of psychologically harmful caregiving so that we can try to make more conclusive statements about how they may affect development. Therefore, we seek to examine inhibitory control as a potential cognitive mechanism that underlies the long-term effects of childhood emotional abuse and emotional neglect. By integrating contemporary cognitive and developmental psychology with Indian philosophical perspectives on the mind, the study aims to uncover how attentional regulation is affected, the extent to which it is and whether these effects vary based on experiences of neglect and abuse.

Primary Objectives

- To examine the association between childhood emotional abuse, emotional neglect and inhibitory control in emerging adults.
- To distinguish the effects of emotional abuse and emotional neglect on inhibitory control rather than treating psychologically harmful caregiving as a single construct.
- To investigate whether emotional abuse and emotional neglect are associated with different patterns of inhibitory control, consistent with contemporary developmental models of threat and deprivation.
- To compare performance on hot (emotionally salient) and cold (emotionally neutral) inhibitory control paradigms in individuals reporting varying levels of childhood emotional abuse and emotional neglect.
- To determine whether emotionally salient distractors differentially influence inhibitory control among individuals with histories of emotional abuse and emotional neglect.

Secondary Objectives

- To evaluate whether inhibitory control provides a useful cognitive mechanism for explaining the heterogeneous findings reported in previous psychologically harmful caregiving research.
- To gather more empirical data to support the psychological maltreatment framework as a model of psychologically harmful caregiving, which is still a relatively understudied aspect of childhood adversity in comparison to physical and sexual abuse.
- To compare evidence from the Indian sociocultural context where emotional abuse and emotional neglect are normalised, underrecognised and highly prevalent, with the existing literature from western societies.
- To integrate Indian philosophical cognitive frameworks, especially the concepts of Manas, Buddhi and Samskara, with contemporary theories of executive functioning and caregiving experiences to encourage collaborative science, the preservation of Indian traditional knowledge and resist dominant western frameworks of knowledge.
- To lay the foundation for future research investigating the different cognitive mechanisms affected by psychologically harmful caregiving and lead to better understanding, assessment, intervention and educational practices surrounding trauma.

Hypotheses

The hypotheses tested in this proposed study are derived from the previously discussed conceptual frameworks and are aimed to empirically evaluate the effects of psychologically harmful caregiving on inhibitory control. Specifically, we intend to look at whether this relationship varies based on the form of psychologically harmful caregiving and the emotional context in which inhibitory control is tested.

Emotional Neglect

H1: Higher levels of childhood emotional neglect, as measured by the Emotional Neglect subscale of the CTQ-SF, will be associated with poorer inhibitory control performance across all emotional distractor conditions (neutral, negative, and positive). Poorer inhibitory control will be operationalized as lower d' score, higher commission and omission error rates, slower mean reaction times, and greater reaction time variability on the Emotional Go/No-Go task.

In the current literature, emotional neglect during childhood is seen as a form of deprivation that has long-term consequences for the development of executive functions. Studies show that neglect is different from other forms of psychologically harmful caregiving as it deprives children of emotional, cognitive stimulation and supportive caregiver experiences that are necessary for the maturation of these higher-order cognitive functions [29,17]. Neglect

primarily disrupts these functions, particularly those involving the PFC. Therefore, children who experience neglect are expected to have impairments in inhibitory control and other functions independent of the emotional context or other details. These functions are highly malleable during childhood and adolescence and depend on caregivers for their normal development. Interventions that enhance cognitive stimulation and caregiver responsiveness have already been found to improve inhibitory control and other executive functions among neglected children [30].

These findings are further explained by the Dimensional Model of Adversity and Psychopathology where emotional neglect is conceptualised as a form of deprivation that limits crucial developmental processes necessary for higher-order cognitive functioning. This understanding is supported by empirical evidence that shows the neurodevelopmental consequences of neglect. Hanson et al. found that children who faced early neglect showed reduced white matter integrity in prefrontal pathways that are important for executive functioning. They also argue that understanding these consequences may help the formulation of targeted interventions to improve cognitive functioning among children who have faced neglect [31].

Similar findings have been reported in longitudinal studies of children exposed to severe early psychosocial deprivation. These studies demonstrate persistent impairments in executive functioning, including inhibitory control, that continue into later childhood and adolescence despite subsequent improvements in caregiving environments [23]. Together, these findings suggest that deprivation experienced during sensitive developmental periods can alter the maturation of executive control systems in enduring ways rather than producing only short-term cognitive difficulties.

Emotional Abuse

H2: Higher levels of childhood emotional abuse, as measured by the Emotional Abuse subscale of the CTQ-SF, will be associated with inhibitory control performance specifically under emotionally salient distractor conditions but not under neutral distractor conditions.

While emotional neglect is viewed as a form of deprivation, emotional abuse is understood as a form of threat. Constant experiences involving criticism, humiliation, rejection, intimidation and verbal abuse may expose children to chronic interpersonal threat. Instead of a general disruption of executive functions and their development, these kinds of experiences recalibrate attentional systems, making them more sensitive to emotionally valent information (especially cues signalling potential danger) [10]. These adaptations may be beneficial during childhood, in threatening situations, but persist into adulthood when they are no longer adaptive.

Therefore, emotional abuse is expected to affect cognitive functioning in emotionally salient contexts rather than produce a general disruption.

Empirical evidence supports this distinction. A study investigating attentional biases in adults with varying childhood caregiving experiences found that emotional abuse was specifically associated with faster detection of negatively conditioned faces, whereas emotional neglect was associated with impaired recognition of familiar faces regardless of emotional content [12]. These findings suggest that emotional abuse preferentially alters the allocation of attention towards emotionally relevant stimuli instead of producing generalized cognitive deficits. Similar patterns have been reported across other facial emotion processing paradigms, where individuals exposed to childhood threat demonstrate heightened vigilance towards negative or threatening information [32,33].

However, only a few studies have specifically investigated if these attentional changes extend to inhibitory control. Individuals need to suppress responses to salient, task-irrelevant stimuli to successfully exercise inhibitory control. Emotional distractors place higher demands on this function, in comparison to neutral stimuli, as we automatically attend to them. Current literature using Emotional Go/No-Go paradigms has shown that psychologically harmful caregiving can influence inhibitory control differently based on emotional context [26]. However, this study treats psychologically harmful caregiving as a single construct and does not distinguish between abuse and neglect nor physical, sexual and emotional maltreatment. The present study addresses this limitation by examining emotional abuse and emotional neglect as separate dimensions of psychological maltreatment. Based on existing theoretical and empirical evidence, it is hypothesized that higher levels of childhood emotional abuse will be associated with altered inhibitory control specifically during emotionally salient distractor conditions, while inhibitory control under neutral conditions will be comparatively less affected.

Differential Effects

H3: The interaction between childhood emotional abuse and emotional distractor condition will be significantly stronger than the corresponding interaction for emotional neglect, indicating that emotional abuse primarily influences inhibitory control in emotionally salient contexts, whereas emotional neglect is associated with generalized inhibitory control performance irrespective of emotional context.

While relatively new theoretical models argue for a distinction between deprivation- and threat-related forms of psychologically harmful caregiving, only a few studies have compared their effects within a single experimental paradigm. These studies either examine psychologically harmful caregiving as a single construct or investigate the subtypes separately from each other, which makes it difficult to determine if the observed effects reflect deeper

variation in the underlying cognitive mechanisms or simply methodological differences across studies. By simultaneously testing emotional abuse and neglect within the same sample and task, the proposed study is able to directly test these competing developmental pathways.

Demonstrating differential patterns would offer empirical support for the Dimensional Model of Adversity and Psychopathology and strengthen the argument that psychological maltreatment should not be conceptualized as a unitary construct. Instead, different forms of psychological maltreatment may influence inhibitory control through distinct cognitive mechanisms, with important implications for both theory and intervention.

Major Research Questions

Even though we already know that psychologically harmful caregiving has been consistently linked with worse cognitive and psychological functioning, there are still pertinent unanswered questions regarding the specific mechanisms through which these long-term effects come about. The overarching question is whether negative caregiving experiences during one's childhood can have lasting impacts on cognitive abilities. This is important not only because it allows us to understand and prevent these negative experiences from happening, but also because it helps us develop targeted intervention plans that address the underlying mechanism that's affected. There are two other questions that are important for the proposed study.

First is whether all kinds of psychologically harmful caregiving are similar in their impacts (especially on executive functioning). Even though there is a significant number of studies exploring the need for a more nuanced understanding of these early life experiences, most research still investigates psychological maltreatment as a unitary construct. This approach may obscure the important differences between the various kinds of childhood adversities and may be the reason for inconsistencies in the current literature. Therefore, it is important to address whether emotional abuse and emotional neglect affect inhibitory control differently or if they produce similar effects.

Second is whether the effects of psychologically harmful caregiving on executive function tasks depend on their emotional context. Though previous studies have looked at the impacts on inhibitory control, they rarely manipulate the emotional contexts of these tasks. However, we must consider that emotional distractors place greater demands on attentional control than neutral stimuli and this might lead to new insights. More importantly, we have seen that contemporary theoretical frameworks suggest that experiences of threat may increase sensitivity to emotional information which may shape the development of these executive functions differently. In the aim of answering these broad questions, here are the specific questions the proposed study aims to answer:

- How do emotional abuse and emotional neglect during childhood differentially influence inhibitory control during early adulthood?
- Can we associate childhood emotional neglect with general inhibitory control deficits that are independent of emotional context?
- Does childhood emotional abuse have unique effects on inhibitory control in the presence of emotionally salient distractors?
- Which behavioral measures from the Go/No-Go task (d' , commission errors, omission errors, reaction time, and reaction time variability) are most strongly associated with emotional abuse and emotional neglect?
- After controlling for other forms of maltreatment and current psychological distress, does childhood psychological maltreatment affect inhibitory control?
- Can we explain the inconsistencies in our current literature by incorporating emotional contexts into inhibitory control paradigms?
- To what extent does studying emotional abuse and emotional neglect independently improve our understanding of psychologically harmful caregiving among Indian emerging adults?

These questions aim to fill the important gaps in the literature by integrating contemporary developmental theories with experimental measures of inhibitory control. They also seek to determine whether separating emotional abuse from emotional neglect offers greater explanatory value than treating psychological maltreatment as a single construct, particularly within the Indian context.

Proposed Research Methodology and Sample Design

Design

The proposed study uses a cross-sectional, correlational research design. We aim to use a within-subjects manipulation of emotional distractors (neutral, negative and positive) and continuous predictor variables (emotional abuse and neglect). Since harmful caregiving experiences, and psychological maltreatment in particular, do not occur in binaries but in a spectrum of severity and frequency, a group comparison approach (maltreated vs non-maltreated) may not be the best fit. Separating maltreatment into groups would discard meaningful variability in severity and frequency and reduce statistical power. A dimensional approach better enables us to account for this pattern while also allowing us to examine emotional abuse and emotional neglect as separable constructs.

Sample

The target population for the proposed study is emerging adults, from age 18 to 25, drawn from Indian undergraduate and postgraduate student populations. During this age range, executive functions reach relative maturity and is usually the time individuals gain increasing social and psychological autonomy from their caregivers. This allows us to examine the long-term impacts of harmful caregiving. A preliminary target sample of approximately 150 participants will be recruited. This is proposed based on previous studies that employ a similar, multiple regression and mixed effects models with anticipated effect size of small to moderate. A formal a priori power analysis will be conducted at the design finalization stage to confirm the precise number of participants needed at 80% power with $\alpha = .05$.

Since childhood emotional abuse and emotional neglect will be measured as continuous individual variables using the Childhood Trauma Questionnaire, there will be no pre-screening of participants based on maltreatment. This also helps prevent ethical and practical obstacles with recruiting participants on the basis of disclosed abuse history. Exclusion criteria will include any self reported diagnosed neurological conditions that are known to affect cognitive functions and visual problems that may interfere with performance on the visual task.

Sampling Procedure

Purposive and snowball sampling methods will be used to recruit participants across universities in the Delhi-NCR region via institutional mailing lists, social media and voluntary sign up. Recruitment invitations will only describe the study in general terms such as “childhood experiences” and “attention” to minimize demand characteristics, social desirability and other biases, especially due to the social stigma and underreporting of psychologically harmful caregiving experiences.

Measures

Childhood Trauma Questionnaire – Short Form (CTQ-SF)

This questionnaire is a 28-item retrospective self-report measure assessing five subtypes of maltreatment (emotional abuse, physical abuse, sexual abuse, emotional neglect, and physical neglect) along with a minimization/denial validity scale. The CTQ-SF is the most widely validated instrument of its kind internationally and allows dimensional scoring, making it well suited to our analytic approach [34]. Physical and sexual abuse subscale scores will be included as covariates to isolate the unique contribution of emotional abuse and emotional neglect.

Emotional Go/No-Go Task

Adapted from paradigms used by Cohen-Gilbert et al. and Demers et al. (2022) , this computerized task requires participants to respond to frequent "Go" letters and withhold responses to infrequent "No-Go" letters, presented over background images that are negative, positive, or neutral in valence (drawn from a standardized, culturally appropriate image set, adapted where necessary from the international affective picture system to ensure relevance to an Indian sample) [21,26]. Three blocks (one per distractor valence) will be presented in counterbalanced order. Primary outcome variables include: d-prime (signal detection sensitivity, combining hit and false alarm rates), commission errors (failures to withhold on No-Go trials), omission errors, and mean reaction time and reaction time variability on correct Go trials. Following recent recommendations, the Go/No-Go task will employ a high proportion of Go trials and short inter stimulus intervals to establish a prepotent motor response. This ensures that successful No-Go trials require genuine response inhibition rather than simple response selection [35].

DASS-21

The Depression, Anxiety and Stress scale is a brief self report assessment of the current mental status of the participants. It will be included to statistically control for the participants' affective state which may possibly influence their overall performance or on the emotionally salient distractor tasks. Current literature suggests that it may be better to use the DASS-21 as a measure of a general distress construct. Bifactor analyses show that a common underlying factor may account for the majority of the variance across items in the scale. Therefore, the DASS-21 provides an efficient assessment of the participants' distress levels for the proposed study while minimising the need to administer multiple assessments for different distressors [36].

Procedure

The participants start by providing informed consent and will continue with a demographic questionnaire, CTQ-SF and DASS-21 through an online survey. After this, they will complete the Go/No-GO task in a controlled lab setting. The experiment will begin with a practice block to ensure that participants comprehend the goal and rules of the task. This will be followed with three counterbalanced experimental blocks for each emotional context. The full session is not expected to make more than 45-60 minutes. Participants will be debriefed at the end with all the relevant details of the study and the lead researcher's contact information. Given the sensitive nature of the study topic, they will be provided with contact details for university counseling services and other mental health helplines as a precautionary step.

Data Analysis

The data from the proposed study will be analysed using linear mixed effects models (LLMs) which is apt for the repeated measures experimental design. Performance on the inhibitory control task (reaction time, accuracy, commission and omission errors) will serve as the dependent variables. The distractor conditions (neutral vs negative vs positive) will be used as the within subject factor. Emotional abuse and emotional neglect scores from the CTQ-SF will be entered as continuous predictors. The interaction effects between the subtypes of psychological maltreatment and distractor conditions will be examined to determine if emotional abuse and emotional neglect differentially affect inhibitory control under different emotional contexts. In order to account for other confounding variables, the analyses will control for other forms of childhood maltreatment, current psychological distress (as measured by DASS-21) and other relevant demographic variables. Significant interaction effects will be further assessed using post-hoc simple effects analyses wherever appropriate. Effect sizes, confidence intervals at 95%, and significance values will be reported.

Ethical Considerations

Prior to the start of data collection, the proposed study will be reviewed and approved by the Institutional Ethics Committee. All participants will provide informed consent only after receiving detailed information on the study procedures, risks and their rights as participants. Their participation will be entirely voluntary and they may choose not to answer any questions or withdraw from the study at any stage without loss of compensation. Given the sensitive nature of the study topic, it will only employ standardised self report measures instead of demanding the participants to disclose their personal experiences in detail in order to minimise potential distress. At the end of the study, all the participants will be debriefed and provided with information on mental health support services regardless of their responses. All the data collected during the study will be anonymised and stored securely (will only be accessed by the research team). Confidentiality and privacy will be maintained throughout the study, and all procedures will adhere to institutional and national ethical guidelines for research involving human participants.

Relevance and Expected Outcomes in Reference to Decolonisation Studies

While psychology aims to understand human behaviour and mind broadly, most of the existing literature stems from research conducted in wealthy, western populations. Henrich, Heine and Norenzayan reported that 96% of the participants in the well reputed journals belonged to Western, Educated, Industrialised, Rich and Democratic (WEIRD) population [37]. This concern is prevalent in developmental psychology where even though researchers are expanding their scope to the underrepresented population, the field assumes the universality and centrality of studies conducted with wealthy, western populations [38]. These assumptions may

lead to culturally specific findings from the WEIRD population being misattributed as a universal process of human development [38] which may reduce the cross cultural validity of psychological theories.

Parenting is central to a child's developmental trajectory and is culturally embedded. In a study on parenting among Indian adolescents, Sharma et al. proposed that it should be understood within its relational and sociocultural ecology [40]. Their findings suggest that parenting styles assume meaning from the socio-cultural environment. By extension, the implications of parenting, including psychological maltreatment, should also be seen from a cultural lens, rather than in isolation. This reflects the need to examine the developmental consequences of childhood psychological maltreatment, including its effects on inhibitory control, within the Indian context.

The proposed study aligns with Durganand Sinha's framework of exogenous indigenisation, which promotes the integration of existing psychological theories within Indian local context, provided it is critically examined and adapted for the non western population [41]. The present study does not assume the universality of the established developmental pathways and relationships, and evaluates the association between childhood psychological maltreatment and inhibitory control within Indian young adults. It also recognises that the experiences and interpretation of emotional abuse and neglect are culturally embedded in norms and parenting style. Behaviours such as harsh criticism, conditional affection and public humiliation, often normalised within the Indian context [42], are also examined in the study for its potential developmental consequences on children.

If the findings converge with the existing literature, they will strengthen the cross cultural validity of the established relationship between childhood psychological maltreatment and inhibitory control while providing empirical evidence in the Indian context. However, if the findings differ, they will indicate the importance of sociocultural context in shaping the developmental consequences of childhood psychologically harmful caregiving. This will help identify the need to refine existing developmental theories, while situating them within Indian social, cultural and philosophical context, which are often not adequately represented within the existing psychological frameworks.

In either case, the study positions the Indian population as contributors to knowledge production, thereby contributing to the decolonisation of developmental psychology. Beyond, the findings within Indian local context, may help psychologists, educators and policy makers to better understand the impact of childhood psychological maltreatment and thereby, may contribute in the development of culturally relevant screening tools, trauma informed interventions, parenting programmes and child welfare policies. The proposed study will also help establish an empirical foundation for future research on childhood psychological

maltreatment and contribute towards developing culturally grounded psychological theory and practice surrounding Indian parenting and child development.

Total Grant Amount Expected for The Study

Head	Amount
Student stipend	₹60,000
Field work	₹60,000
Research equipment & study materials	₹60,000
Contingency	₹20,000
Grand Total	₹2,00,000

Justification

Student Stipend (₹60,000)

The proposed study will be conducted by one Lead Researcher and one Researcher-1 over a period of six months. In line with the ICSSR guidelines, both researchers will receive a monthly stipend of ₹5,000. This will help support their sustained involvement in the study and heavily motivate them to put in the most possible effort and commit a substantial amount of time throughout the project.

Field Work (₹60,000)

Fieldwork expenses will primarily support participant recruitment and data collection across universities in the Delhi-NCR region. A major component of this budget is participant compensation (approximately ₹100–150 per participant) to acknowledge the time spent completing the questionnaires and laboratory-based Emotional Go/No-Go task. The remaining amount will cover local travel for researchers, recruitment activities, printing of informed consent forms and study materials, and other logistical expenses incurred during data collection.

Research Equipment and Study Materials (₹60,000)

This budget head will support the purchase of equipment and materials required for the successful execution of the study. These include a laptop and peripheral devices for administering computerized cognitive tasks, secure external storage devices for research data,

books and reference materials, stationery, printing costs, and other study-related resources. Any equipment purchased under the project will remain the property of the affiliating institution in accordance with ICSSR guidelines.

Contingency (₹20,000)

This contingency fund will help meet any unforeseen research related expenses that may come up during the project. This may include any conference fee, additional printing or stationary charges, transportation costs for researchers and study participants, and other incidental expenditures that we need to account for to complete the research on time. The proposed budget has been formulated following the ICSSR guidelines and is designed keeping in mind the importance of prudent utilization of grant funds. This will help ensure the effective completion of all research tasks without compromising on quality. This support will enable the research team to produce their best, methodologically rigorous work with a strong sample size and high quality data. By addressing an important yet understudied area of childhood maltreatment in the Indian context, the proposed research can inform future research, trauma informed educational practices, mental health interventions and more. The financial assistance provided by ICSSR will play an extremely crucial role in supporting meaningful scientific inquiry and researchers who look forward to contributing more to the Indian scientific community.

Milestones

Timeline	Goals
Months 1 -2	Finalization of the research protocol and study materials; Institutional Ethics Committee approval; programming and pilot testing of the Emotional Go/No-Go task; procurement of research equipment and materials; preparation of participant recruitment materials.
Months 3 - 4	Participant recruitment across universities in the Delhi-NCR region; administration of informed consent, questionnaires (CTQ-SF and DASS-21), and Emotional Go/No-Go task; ongoing monitoring of data quality and secure data management.
Months 5 - 6	Completion of recruiting participants and collecting data; data cleaning, coding, and preliminary statistical analyses; verification of data quality and preparation for more advanced analysis.

Months 7 - 8	Final statistical analyses using linear mixed-effects models; interpretation of findings; preparation of the final project report; discussing findings through conference presentations and preparation of a manuscript for publication in a peer-reviewed journal.
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